SA
Principles

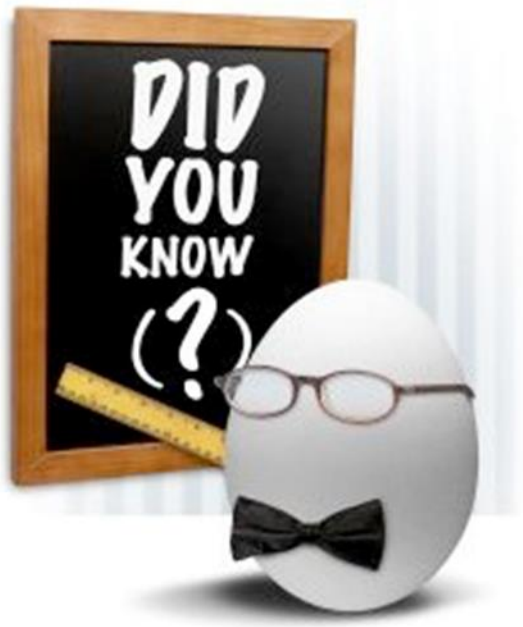
GRASP

Knowing and Doing Responsibilities

Responsibility-Driven Design (RDD)

- "**Doing**" Responsibilities
  - Create another object
  - Perform a calculation
  - Initiate an action in an object
  - Control/coordinate activities of objects

- "**Knowing**" Responsibilities
  - Knowing it's own encapsulated data
  - Knowing about other/related objects
  - Knowing things it can derive or calculate



General Responsibility Assignment Software Patterns (GRASP)

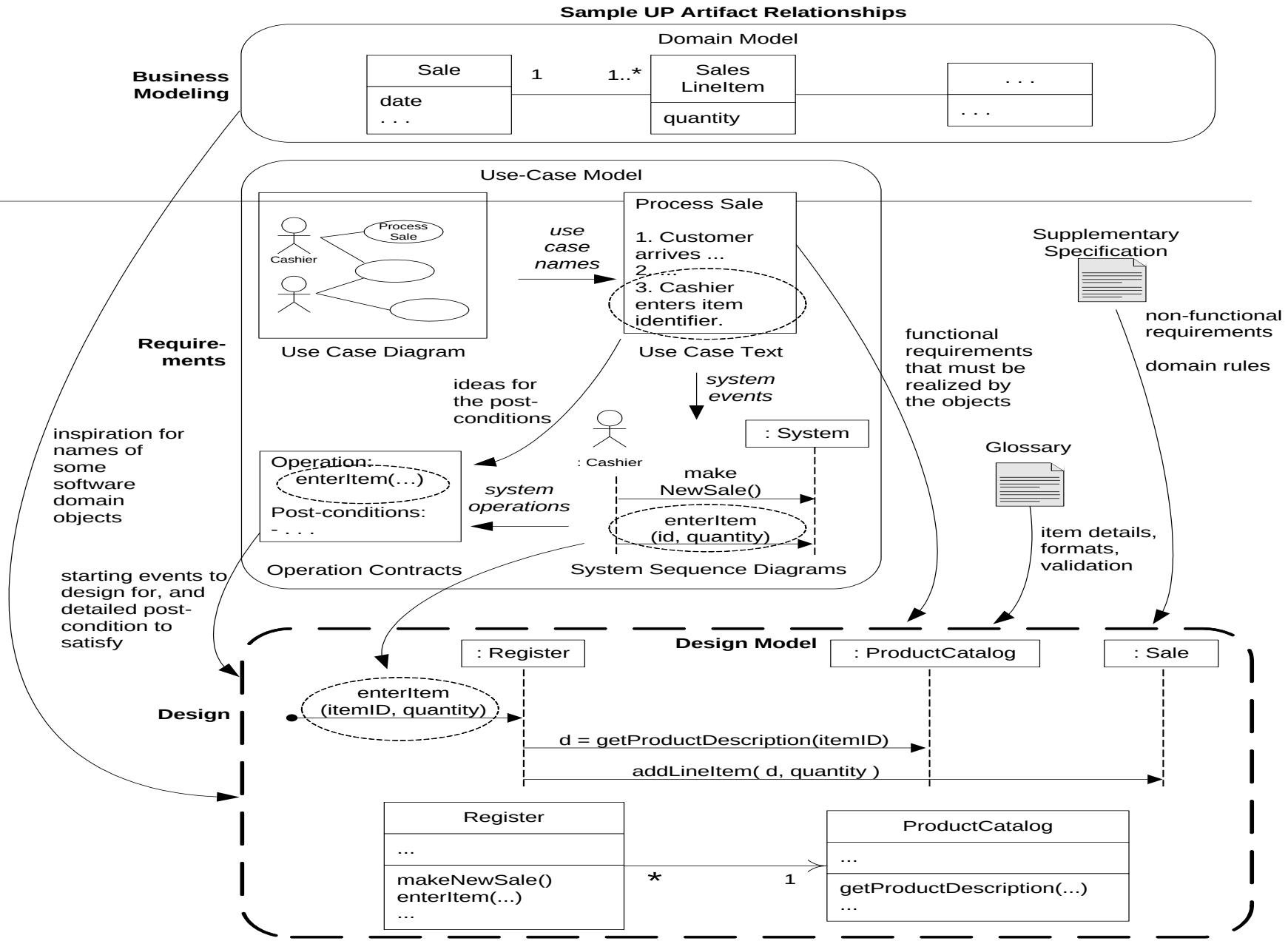
- General Responsibility Assignment Software Patterns (or Principles)
  - A set of patterns for assigning responsibilities to software objects

- Factors / principles

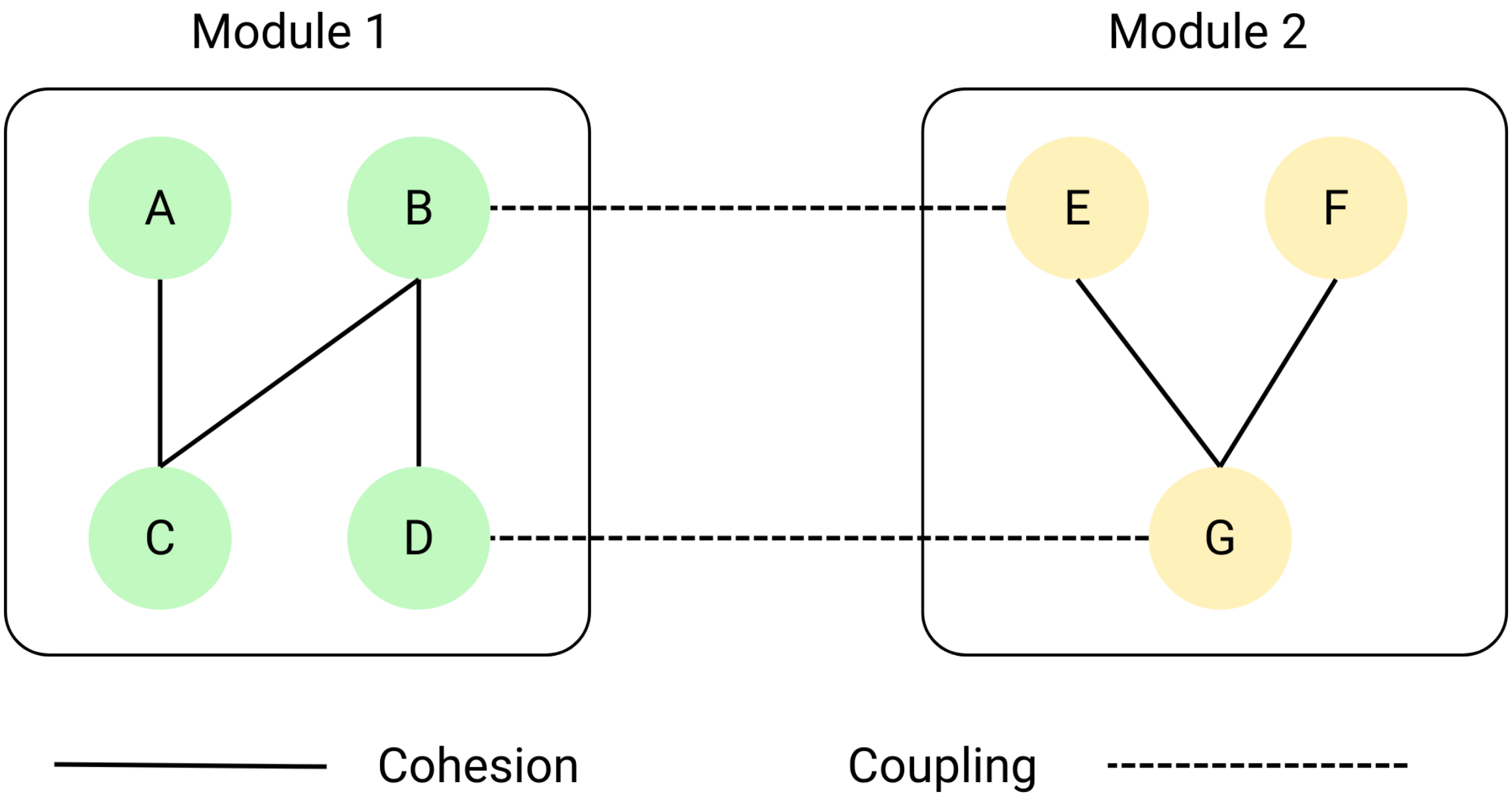
Creator	**Controller**	**Pure Fabrication**
Information Expert	**High Cohesion**	**Indirection**
Low Coupling	**Polymorphism**	**Protected Variations**

Ex.

Design Class	Responsibility
Sale	Knows sale total
SalesLineItem	Knows line item subtotal
ProductSpecification	Knows product price

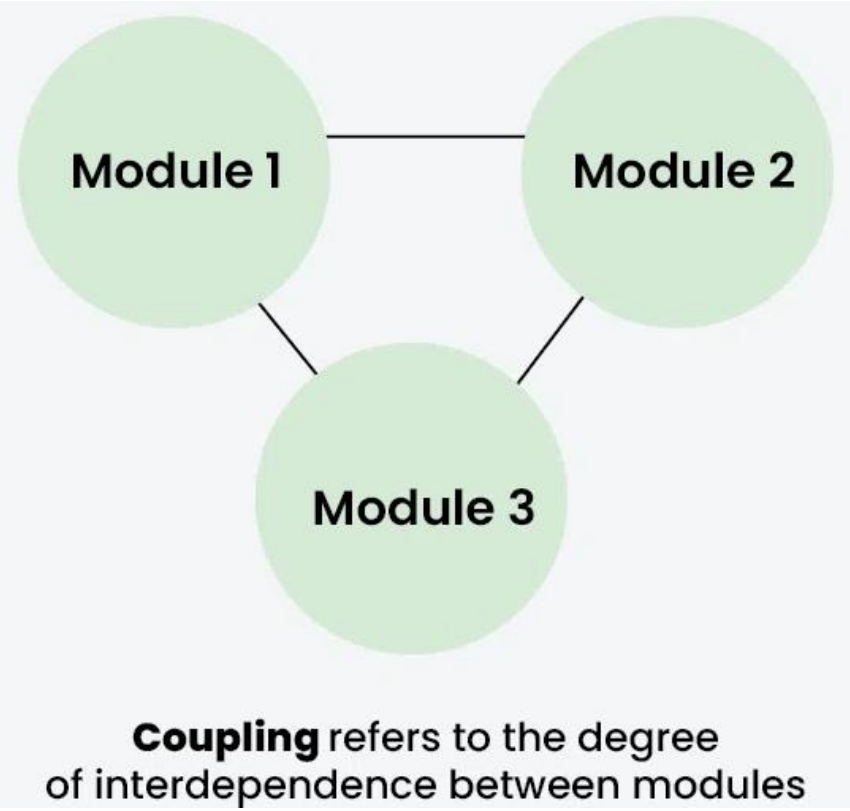


Coupling and Cohesion

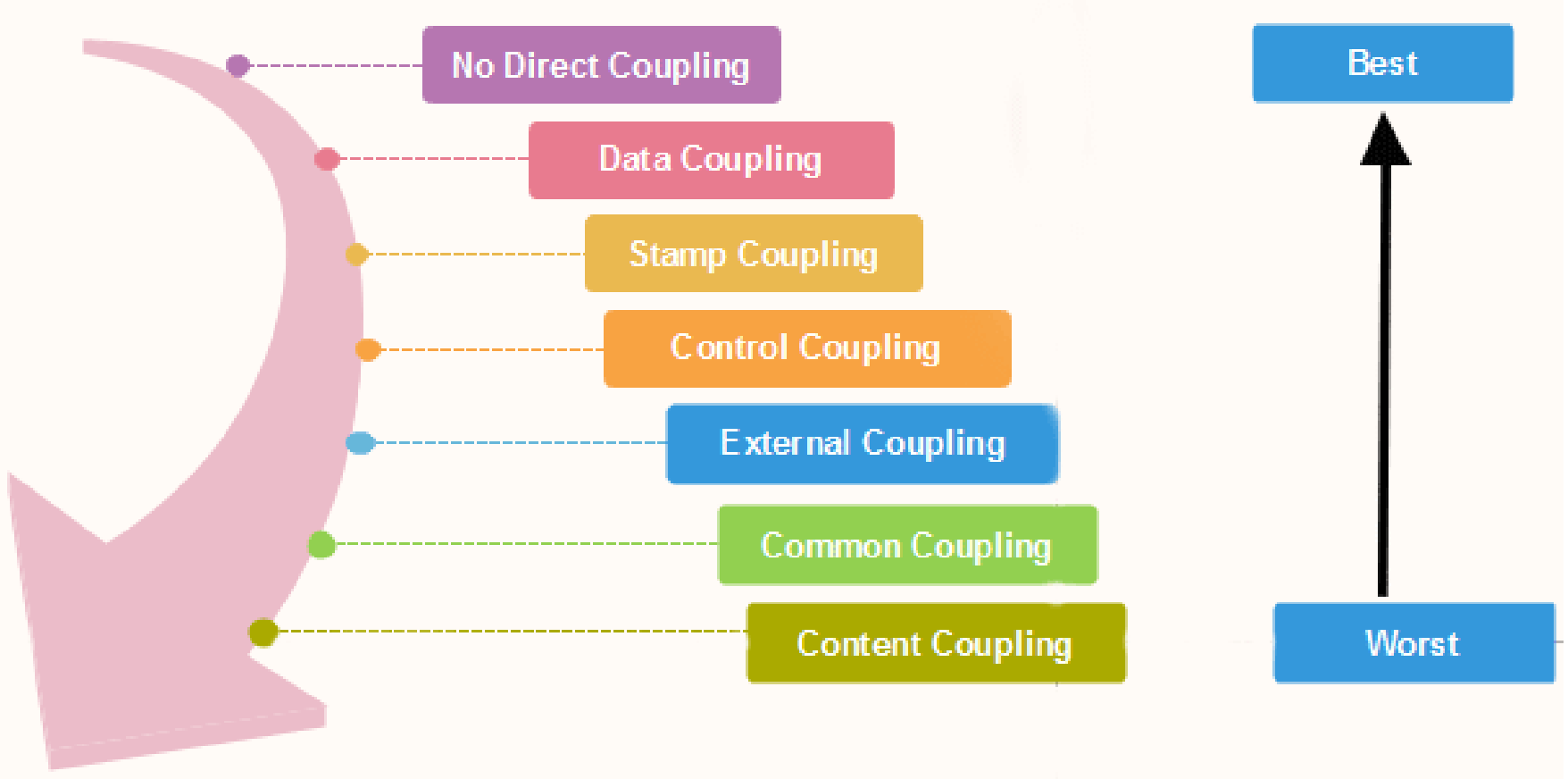

Module 1
Module 2
A
B
C
D
E
F
G
Cohesion
Coupling

Coupling

- Coupling refers to the degree of interdependence between software modules.
  - High coupling means that modules are closely connected and changes in one module may affect other modules.
  - Low coupling means that modules are independent, and changes in one module have little impact on other modules.

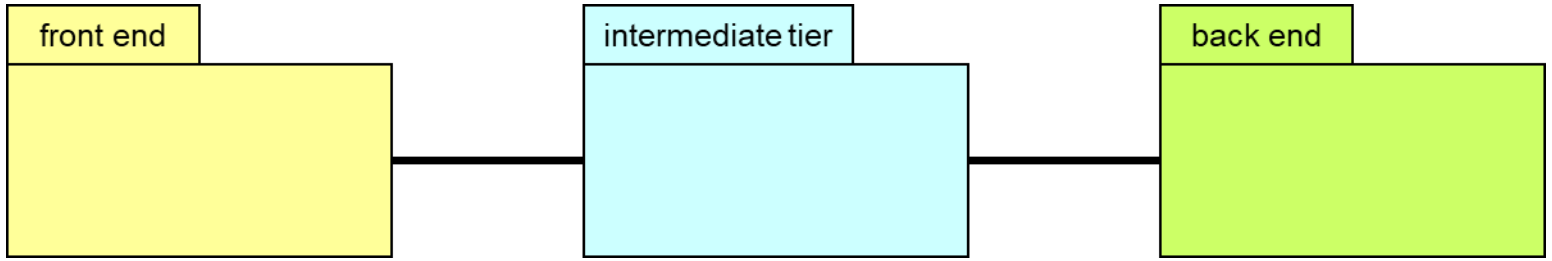


Types of Coupling



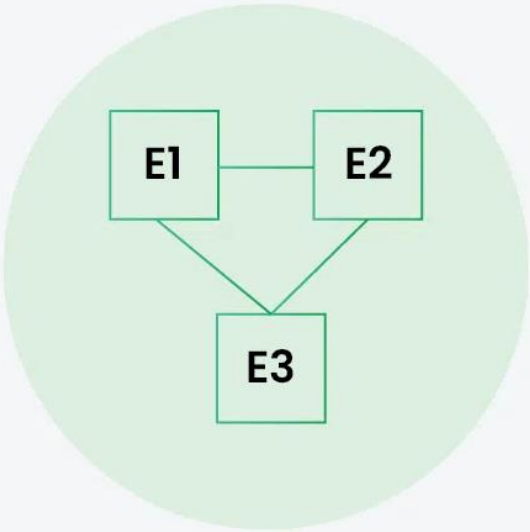
Coupling – metrics

- Coupling refers to the number of associations (dependencies) between two subsystems.
  - **Interaction** coupling is a measure of the number of different messages an object sends to other objects and number of passed parameters with these messages.
  - **Inheritance** coupling describes the degree to which a subclass actually needs the features in inherits from it's abase class.



Cohesion

- Cohesion refers to the degree to which elements within a module work together to fulfill a single, well-defined purpose.
  - High cohesion means that elements are closely related and focused on a single purpose,
  - while low cohesion means that elements are loosely related and serve multiple purposes.



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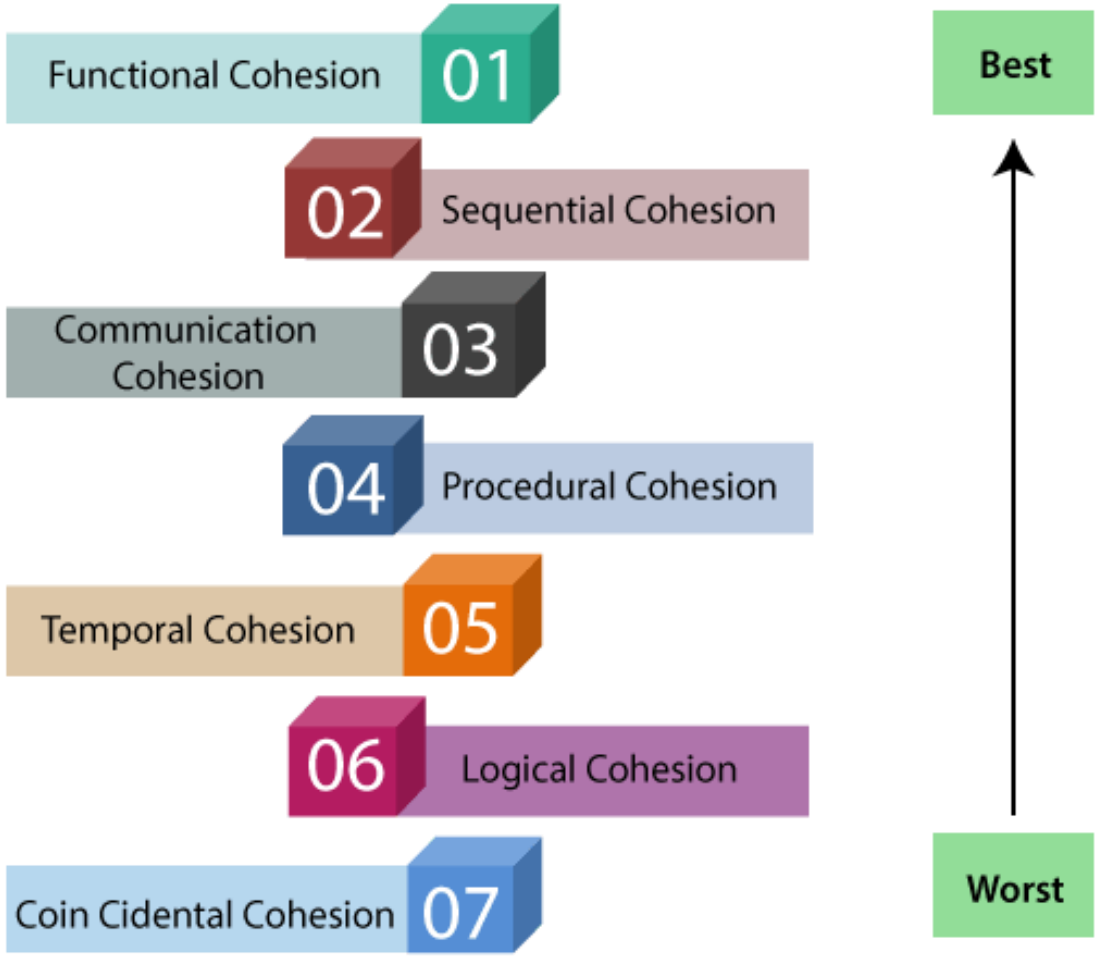
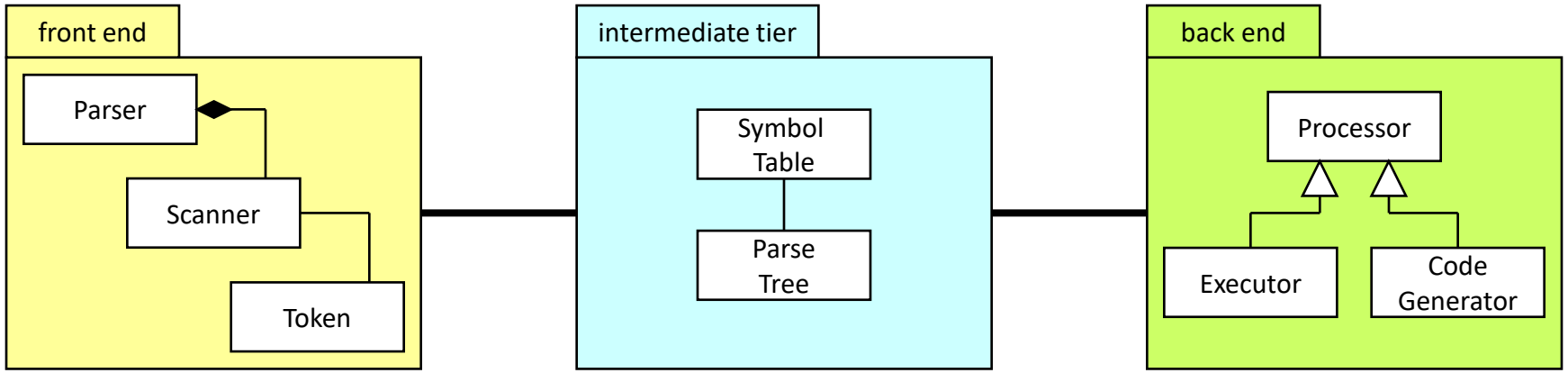
Types of Cohesion



Cohesion – metric

- Cohesion is the number of associations (dependencies) within a subsystem
  - **Operation** cohesion measures the degree to which an operation focuses on a single functional requirement.
  - **Class** Cohesion measures the degree to which a class focuses on a single requirement.



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